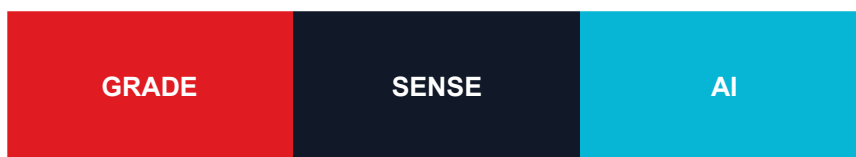


GradeSense Technical Documentation

Architecture, intelligence workflows, explainability, deployment, and results



Explainable predictions, constraint-aware recommendations, and real-time operations for safer, faster paper grade transitions.

Prepared from the validated GradeSense v0.2.0 source repository and live application.

Executive Summary

GradeSense | Honeywell Hackathon Submission

GradeSenseAI is an operator-facing industrial intelligence platform for paper-mill grade transitions. It combines snapshot prediction, sequential basis-weight forecasting, explainable AI, safe intervention simulation, persistent recommendations, alerts, and operational governance in one deployable system.

The platform is advisory by design: every action is constraint-checked, model-simulated, ranked by expected improvement, explained in operational language, and preserved for later effectiveness analysis.

- Decision support spans live sensing, inference, recommendation, operator action, and outcome learning.
- REST and WebSocket interfaces support both deliberate workflows and continuous streaming.
- Immutable model artifacts, audit logs, health checks, and registry controls support production readiness.

Problem Statement

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Paper grade transitions are multivariable, time-dependent processes. Operators must stabilize basis weight and quality while machine speed, moisture, stock flow, steam pressure, temperature, pulp consistency, and grade-specific targets change together.

The objective is to reduce off-spec production and stabilization time without replacing operator authority.

Existing Challenges

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- Delayed detection: conventional thresholds react after deviation is visible.
- Fragmented context: sensor trends, model confidence, and historical outcomes live in separate views.
- Unsafe optimization: an apparently beneficial set-point change can violate process constraints.
- Low trust: black-box predictions do not explain why risk changed or what to do next.
- Weak learning loop: accepted recommendations and observed outcomes are rarely measured together.

Proposed Solution

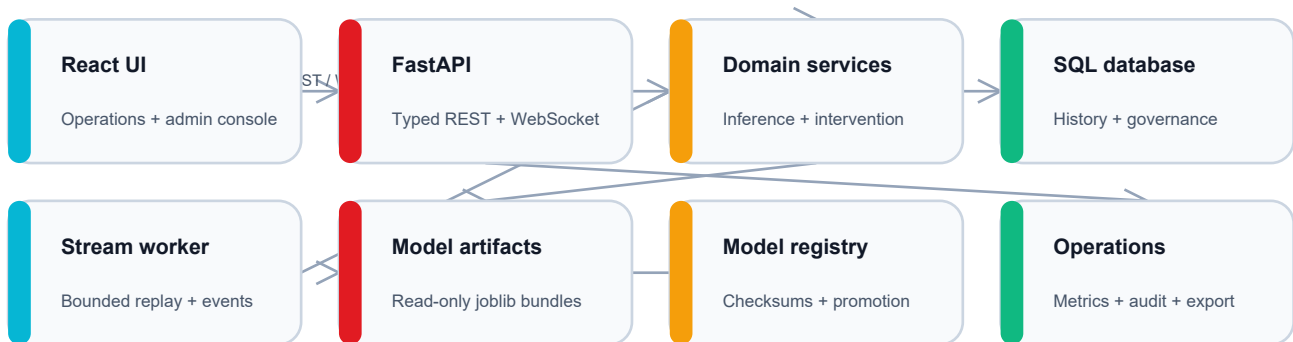
GradeSense | Honeywell Hackathon Submission

GradeSense fuses typed FastAPI services, a React operations console, persisted operational history, and two model families. Snapshot intelligence estimates current transition quality, off-spec risk, and stabilization; sequential forecasting projects basis-weight trajectory and detects specification crossing.

- Generate only feasible intervention candidates.
- Simulate each candidate with the active forecast model.
- Rank improvements deterministically by crossing risk, peak deviation, and stabilization.
- Expose feature-level reasons and confidence instead of opaque scores.
- Stream predictions, alerts, drift, and recommendations to the dashboard.

System Architecture

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ML Pipeline

GradeSense | Honeywell Hackathon Submission

The snapshot pipeline validates and preprocesses a transition state, resolves the active model, produces quality/risk/stabilization outputs, and attaches feature-level explanations. The sequential pipeline enforces transition boundaries, builds lag/rolling/rate features, executes direct-horizon regressors, estimates residual intervals, and monitors the $\pm 2.5\%$ specification band.

- Direct-horizon forecasting avoids recursive error accumulation.
- Windows never cross transition IDs, protecting temporal validity.
- Validation residuals become prediction intervals.
- Model and feature-schema checksums make inference reproducible.

Recommendation Engine

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The intervention engine creates one-to-four-variable candidate adjustments from operationally meaningful ranges. A constraint engine rejects unsafe combinations before inference. Surviving candidates are simulated against a persisted baseline and ranked only when they improve the forecast.

- Candidate generation: single-variable and coordinated multivariable actions.
- Safety gate: bounds, change magnitude, and incompatible combinations.
- Benefit gate: crossing avoidance, lower peak deviation, or faster stabilization.
- Lifecycle: proposed, accepted, rejected, modified, delayed, applied, evaluated.

Explainable AI

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GradeSense turns model evidence into operator-facing reasoning. Snapshot explanations surface the most influential variables and their direction. Forecast recommendations state the proposed set-point changes, why they

are safe, which risk measure improves, and the expected operational effect.

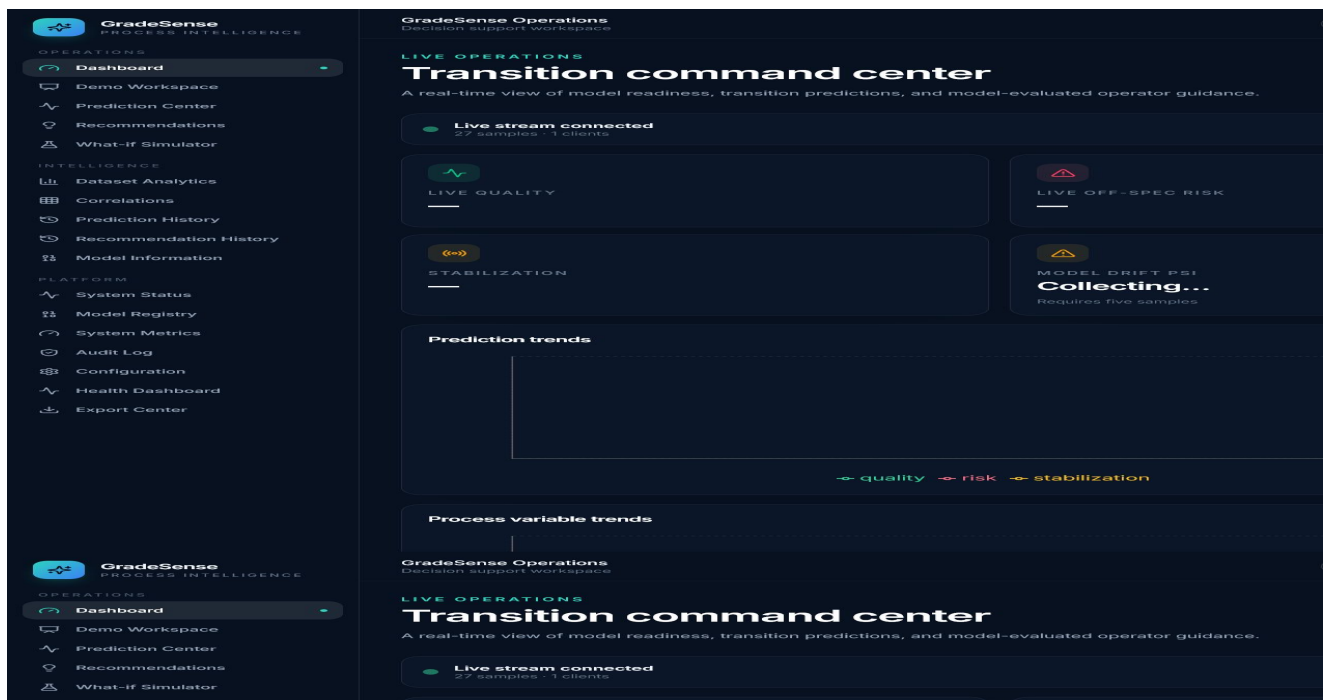
Confidence, model version, data recency, and drift are visible so the operator can judge when the model should carry less weight.

Dashboard

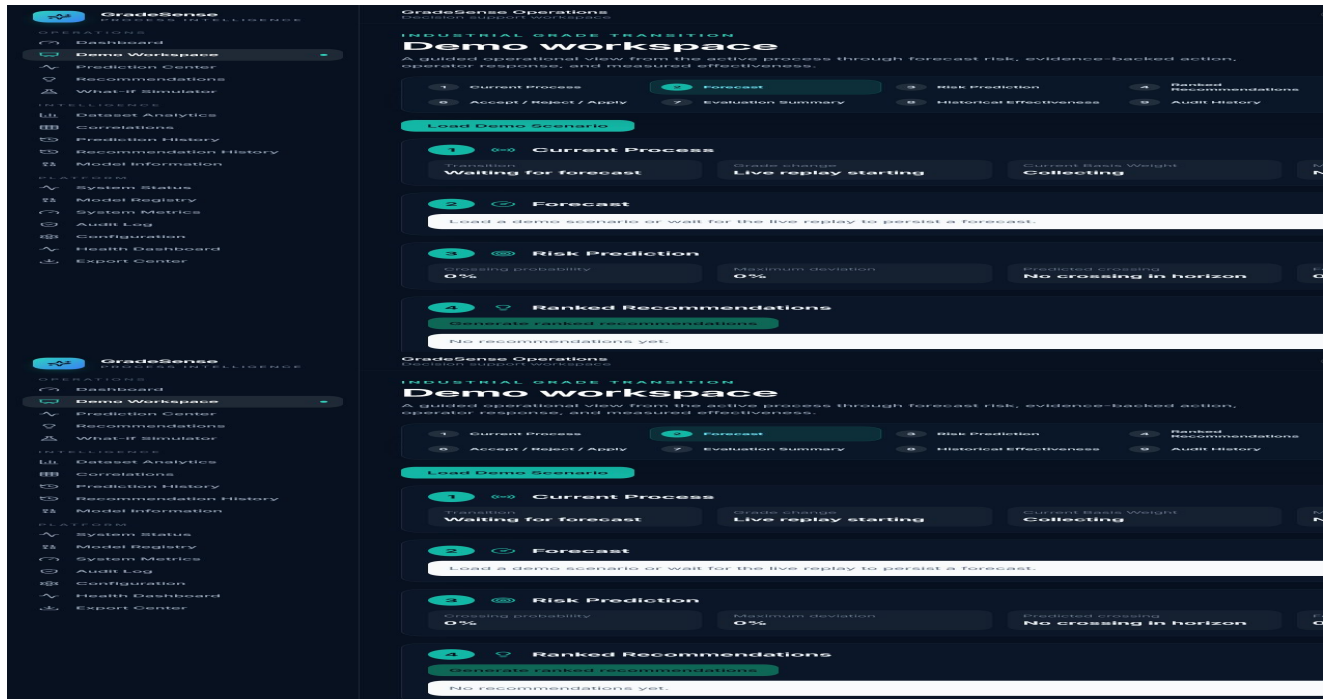
GradeSense | Honeywell Hackathon Submission

The responsive dashboard provides a transition command center, live charts, rolling metrics, alerts, prediction and recommendation workflows, analytics, searchable history, model registry, health, audit, configuration, and export controls.

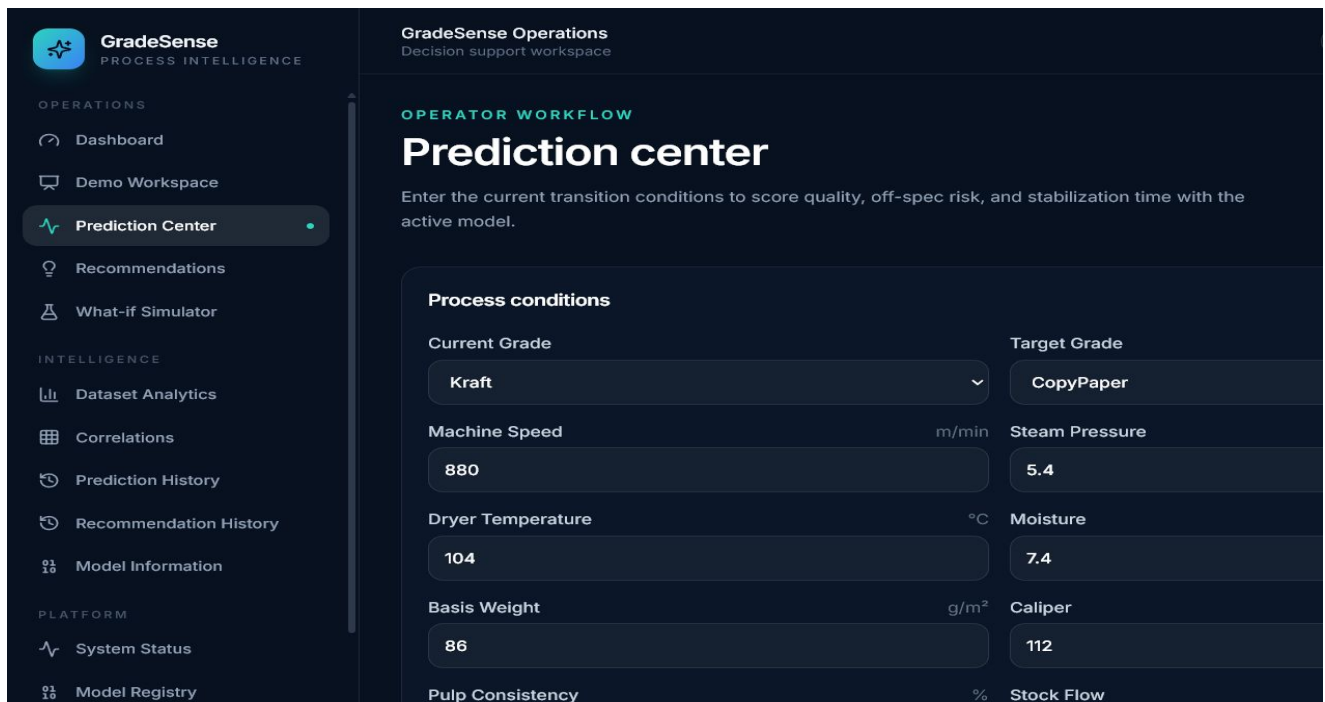
Home



Live Dashboard



Prediction



Recommendation

GradeSense
PROCESS INTELLIGENCE

OPERATIONS

Dashboard

Demo Workspace

Prediction Center

Recommendations

What-if Simulator

INTELLIGENCE

Dataset Analytics

Correlations

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Recommendation History

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PLATFORM

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Model Registry

System Metrics

Audit Log

Configuration

Health Dashboard

Export Center

Intelligence online

Industrial AI platform - v0.2.0

GradeSense Operations
Decision support workspace

System ready

⚙️

GS

DECISION SUPPORT

Recommendation center

Evaluate targeted process interventions against the active model before changing operating conditions.

Transition conditions

Current Grade

Kraft

Machine Speed

880

Dryer Temperature

104

Basis Weight

86

Pulp Consistency

3.5

Refining Energy

160

Reel Tension

5.2

Humidity

72

Target Grade

CopyPaper

Steam Pressure

5.4

Moisture

7.4

Caliper

112

Stock Flow

3400

Headbox Pressure

3.8

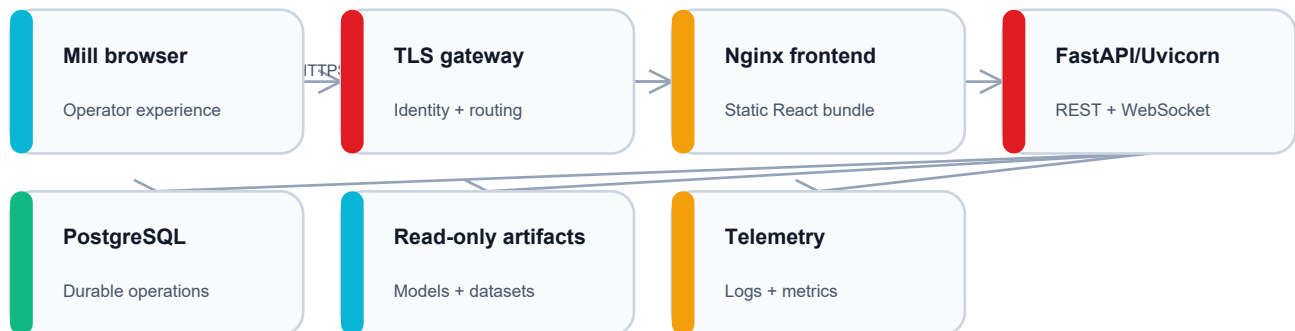
Ambient Temperature

30

Generate recommendations

Deployment Architecture

GradeSense | Honeywell Hackathon Submission



API Overview

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- System: identity, health, version, OpenAPI.
- Intelligence: predict, recommend, correlations, dataset, model, history.
- Forecasting: trajectory, history, simulation, relationship discovery.
- Interventions: ranked actions, decisions, outcomes, effectiveness.
- Real time: alerts, feedback, stream state, rolling/live metrics, drift, WebSockets.
- Administration: registry, configuration, metrics, health, audit, exports.

Technologies

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- Backend: Python, FastAPI, Pydantic, SQLAlchemy, Alembic, Uvicorn.
- ML: scikit-learn-compatible joblib artifacts, feature engineering, direct-horizon regression.
- Frontend: React 19, TypeScript, Vite, Tailwind CSS, TanStack Query, Recharts, Zustand.
- Data and deployment: PostgreSQL/SQLite, Docker Compose, Nginx, Render blueprint.
- Quality: pytest, Vitest, Ruff, ESLint, health checks, structured logs.

Results

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The implemented submission demonstrates a complete inference-to-action loop rather than a standalone model. Automated test suites cover system APIs, real-time behavior, forecasting, recommendations, interventions, datasets, production readiness, and frontend workflows.

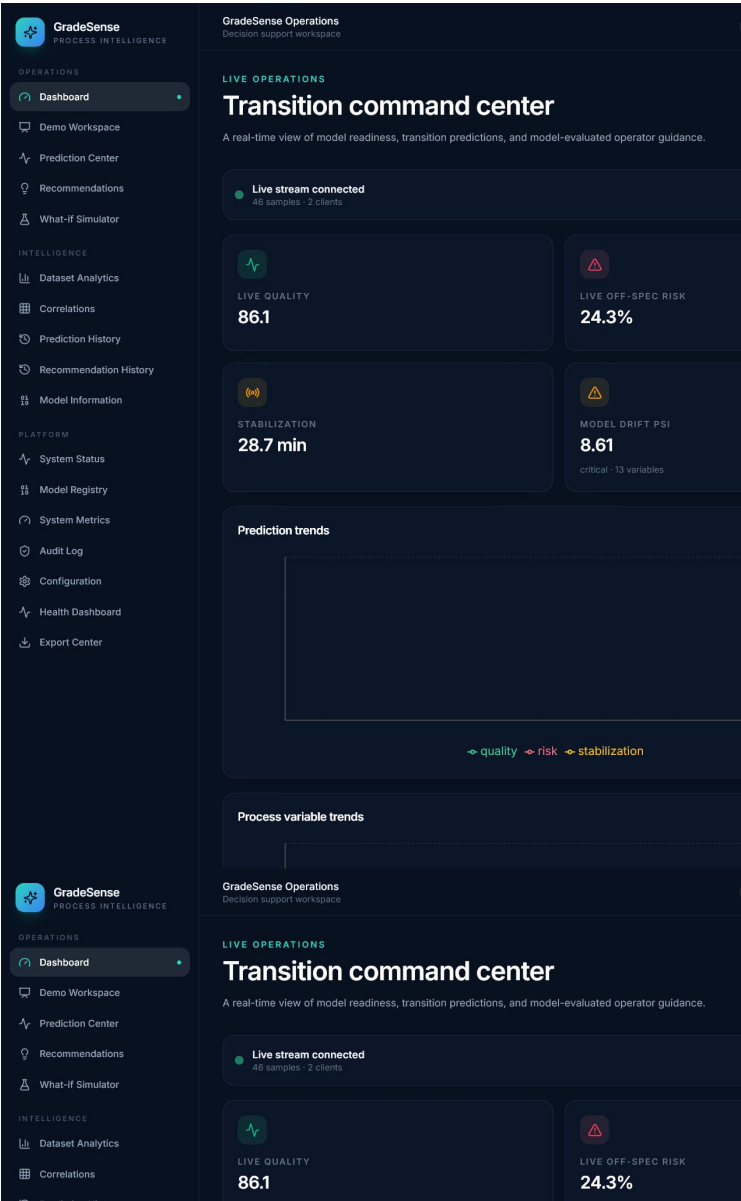
Repository evidence supports the architectural and functional results; the package intentionally avoids inventing plant ROI or model-accuracy figures not present in validated artifacts.

Future Scope

GradeSense | Honeywell Hackathon Submission

- Pilot with live PLC/SCADA integration and plant-specific calibration.
- Add authenticated roles and an identity-aware gateway.
- Move streaming fan-out and rate limits to shared infrastructure for multi-replica scale.
- Introduce governed online learning after sufficient evaluated outcomes.
- Add fleet-level benchmarking, energy optimization, and prescriptive scheduling.

Route: /dashboard · Source: https://gradesense-4weh.onrender.com



Route: /recommendations · Source: https://gradesense-4weh.onrender.com

GradeSense

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%

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kWh/t

Headbox Pressure

3.8

bar

Reel Tension

5.2

kN/m

Ambient Temperature

30

°C

Humidity

72

%

Generate recommendations

Route: /dashboard · Live alerts · Source: https://gradesense-4weh.onrender.com

 **GradeSense**
PROCESS INTELLIGENCE

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GS

Caliper

103.77

Pulp Consistency

3.41

Stock Flow

3,556.96

Refining Energy

154.2

Headbox Pressure

3.46

Reel Tension

3.04

Ambient Temperature

24.01

Humidity

51.83

Live alerts

No live alerts have been raised.

Rolling metrics

Window

1 minute

Predictions

7

Average quality

90.9

Average risk

16.3%

Recommendations

28

Alerts

0

Alert timeline

Recommendation and alert frequency



Route: /analytics · Source: <https://gradesense-4weh.onrender.com>

[illegible]

